

DO NOT ADD OR CREATE ANY NEW CELLS

DO NOT ADD OR CREATE ANY NEW CELLS

DO NOT ADD OR CREATE ANY NEW CELLS

DO NOT ADD OR CREATE ANY NEW CELLS

YOU HAVE TO RUN ALL THESE CELLS BEFORE MOVING ON TO YOUR QUESTIONS. IF YOU ENCOUNTER ANY ERRORS JUST RESTART THE KERNEL

```
In [1]: #Do not modify this cell
#Run this Cell
import pandas as pd
%load_ext sql

In [3]: q1 = pd.read_csv('q1.csv', sep = '\t')
q2 = pd.read_csv('q2.csv', sep = '\t')
q3 = pd.read_csv('q3.csv', sep = '\t')
q4 = pd.read_csv('q4.csv', sep = '\t')
q5 = pd.read_csv('q5.csv', sep = '\t')

In [5]: %sql sqlite:///bookstore.sqlite
```

Here are your questions

You are the database administrator for an online bookstore. The bookstore database has several tables that store information about books, authors, customers, orders, and order items. Your task is to create a set of SQL queries to extract useful information from this database.

- The name of the table are authors, books, customers, order_items, orders, reviews
- Always start your answers with this line `%%sql result <<`. Each question has its own format. For example, Question One would be `%%sql q1_result <<` and so on like that.
- Then start writing your query on the next line for example, Question 2 would be

`%%sql q2_result << select * from orders ```

- If you do not follow these instructions your score will not count.

1. Retrieve a list of the top 5 most popular books based on the number of times they have been ordered. Also order by author name in descending order

```
In [7]: #Run this cell to see the expected output
#do not modify this cell
print('This is the expected output')
q1
```

This is the expected output

	title	author_name	order_count
0	1984	George Orwell	4
1	Adventures of Huckleberry Finn	Mark Twain	3
2	Pride and Prejudice	Jane Austen	3
3	The Hobbit	J.R.R. Tolkien	3
4	Harry Potter and the Philosophers Stone	J.K. Rowling	3

- **Your column names must match the expected output**
- **clear everything in the cell below and write your sql query there.**
- Start your answer by typing this exactly as the first line.

`%%sql q1_result <<`

```
In [9]: %%sql q1_result <<
SELECT b.title, a.name AS author_name, COUNT(oi.order_id) AS order_count
FROM books b
JOIN authors a
ON b.author_id = a.author_id
JOIN order_items oi
ON b.book_id = oi.book_id
GROUP BY b.title, author_name
ORDER BY order_count DESC, author_name DESC
LIMIT 5;

* sqlite:///bookstore.sqlite
Done.
Returning data to local variable q1_result
```

```
In [11]: #Run this cell to check your output
#do not modify this cell
print('This is the output of your query')
q1_result.DataFrame()
```

This is the output of your query

Out[11]:

	title	author_name	order_count
0	1984	George Orwell	4
1	Adventures of Huckleberry Finn	Mark Twain	3
2	Pride and Prejudice	Jane Austen	3
3	The Hobbit	J.R.R. Tolkien	3
4	Harry Potter and the Philosophers Stone	J.K. Rowling	3

```
In [13]: #do not modify this cell
assert q1_result.DataFrame().equals(q1)
```

2. Calculate the total revenue generated by each author from the sales of their books.

```
In [15]: #Run this cell to see the expected output
#do not modify this cell
print('This is the expected output')
q2['total_revenue'] = q2['total_revenue'].apply(lambda x: round(x, 2) )
q2
```

This is the expected output

Out[15]:

	author_name	total_revenue
0	George Orwell	256.83
1	J.K. Rowling	213.88
2	Jane Austen	171.92
3	J.R.R. Tolkien	47.97
4	Mark Twain	26.97

- Your column names must match the expected output
- clear everything in the cell below and write your sql query there.
- Start your answer by typing this exactly as the first line.

%%sql q2_result <<

```
In [17]: %%sql q2_result <<
SELECT a.name AS author_name, SUM(b.price * oi.quantity) AS total_revenue
FROM authors a
JOIN books b
ON a.author_id = b.author_id
JOIN order_items oi
ON b.book_id = oi.book_id
GROUP BY author_name
ORDER BY total_revenue DESC;

* sqlite:///bookstore.sqlite
Done.
Returning data to local variable q2_result
```

```
In [19]: #Run this cell to check your output
#do not modify this cell
print('This is the output of your query')
q2result = q2_result.DataFrame()
q2result['total_revenue'] = q2result['total_revenue'].apply(lambda x: round(x, 2) )
q2result
```

This is the output of your query

Out[19]:

	author_name	total_revenue
0	George Orwell	256.83
1	J.K. Rowling	213.88
2	Jane Austen	171.92
3	J.R.R. Tolkien	47.97
4	Mark Twain	26.97

```
In [21]: #do not modify this cell
assert q2result.equals(q2)
```

3. Identify the top 5 customers who have placed the most orders. Also order by customer name in descending order

```
In [23]: #Run this cell to see the expected output
#do not modify this cell
print('This is the expected output')
q3
```

This is the expected output

Out[23]:

	customer_name	order_count
0	David Wilson	6
1	Alice Johnson	6
2	Eva Martinez	5
3	Charlie Brown	4
4	Bob Smith	4

- **Your column names must match the expected output**
- **clear everything in the cell below and write your sql query there.**
- Start your answer by typing this exactly as the first line.

%%sql q3_result <<

In [25]:

```
%%sql q3_result <<
SELECT c.name AS customer_name, COUNT(order_id) AS order_count
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY customer_name
ORDER BY order_count DESC;

* sqlite:///bookstore.sqlite
Done.
Returning data to local variable q3_result
```

In [27]:

```
#Run this cell to check your output
#do not modify this cell
print('This is the output of your query')
q3_result.DataFrame()
```

This is the output of your query

Out[27]:

	customer_name	order_count
0	David Wilson	6
1	Alice Johnson	6
2	Eva Martinez	5
3	Charlie Brown	4
4	Bob Smith	4

In [29]:

```
#do not modify this cell
assert q3_result.DataFrame().equals(q3)
```

4. Find all orders that contain book with book_id = 1.

In [31]:

```
#Run this cell to check your output
#do not modify this cell
print('This is the expected output')
q4
```

This is the expected output

Out[31]:

	order_id	customer_name	order_date
0	1	Alice Johnson	2023-05-01
1	5	Eva Martinez	2023-05-05
2	13	Eva Martinez	2023-05-15
3	21	Charlie Brown	2023-06-20

- **Your column names must match the expected output**
- **clear everything in the cell below and write your sql query there.**
- Start your answer by typing this exactly as the first line.

%%sql q4_result <<

In [33]:

```
%%sql q4_result <<
SELECT o.order_id, c.name AS customer_name, o.order_date
FROM orders o
JOIN customers c
ON o.customer_id = c.customer_id
JOIN order_items oi
ON o.order_id = oi.order_id
WHERE book_id = 1;

* sqlite:///bookstore.sqlite
Done.
Returning data to local variable q4_result
```

```
In [35]: #Run this cell to check your output
#do not modify this cell
print('This is the output of your query')
q4_result.DataFrame()
```

This is the output of your query

Out[35]:

	order_id	customer_name	order_date
0	1	Alice Johnson	2023-05-01
1	5	Eva Martinez	2023-05-05
2	13	Eva Martinez	2023-05-15
3	21	Charlie Brown	2023-06-20

```
In [37]: #do not modify this cell
assert q4_result.DataFrame().equals(q4)
```

5. Select detailed review information of the Least 5 Books based on ratings, including book title, author name, review rating, and review text.

```
In [39]: #Run this cell to see the expected output
#do not modify this cell
print('This is the expected output')
q5
```

This is the expected output

Out[39]:

	review_id	book_title	author_name	rating	review_text
0	1	1984	George Orwell	5	A timeless masterpiece.
1	2	Pride and Prejudice	Jane Austen	4	A wonderful love story.
2	3	Harry Potter and the Philosophers Stone	J.K. Rowling	5	An incredible journey of magic and friendship.
3	4	Adventures of Huckleberry Finn	Mark Twain	4	A fascinating adventure.
4	5	The Hobbit	J.R.R. Tolkien	5	An epic tale of heroism.

- Your column names must match the expected output
- clear everything in the cell below and write your sql query there.
- Start your answer by typing this exactly.

```
%%sql q5_result <<
```

```
In [41]: %%sql q5_result <<
SELECT r.review_id, b.title AS book_title, a.name AS author_name, r.rating, r.review_text
FROM reviews r
JOIN books b
ON r.book_id = b.book_id
JOIN authors a
ON b.author_id = a.author_id
LIMIT 5;

* sqlite:///bookstore.sqlite
Done.
Returning data to local variable q5_result
```

```
In [43]: #Run this cell to check your output
#do not modify this cell
print('This is the output of your query')
q5_result.DataFrame()
```

This is the output of your query

Out[43]:

	review_id	book_title	author_name	rating	review_text
0	1	1984	George Orwell	5	A timeless masterpiece.
1	2	Pride and Prejudice	Jane Austen	4	A wonderful love story.
2	3	Harry Potter and the Philosophers Stone	J.K. Rowling	5	An incredible journey of magic and friendship.
3	4	Adventures of Huckleberry Finn	Mark Twain	4	A fascinating adventure.
4	5	The Hobbit	J.R.R. Tolkien	5	An epic tale of heroism.

```
In [45]: #do not modify this cell
assert q5_result.DataFrame().equals(q5)
```